

CHAPTER 2

REQUIREMENT ANALYSIS

2.1 Introduction

Requirement analysis is one of the most important phases in software development. It identifies the functional and non-functional requirements of the EduGenie Learning Assistant before implementation. A clear understanding of requirements helps build a reliable, efficient, and user-friendly system.

2.2 Functional Requirements

The system allows users to ask academic questions, obtain AI-generated explanations, generate quizzes, summarize educational content, and receive personalized learning path recommendations. It provides a web-based interface, processes requests using FastAPI, and integrates Google Gemini and LaMini-Flan-T5 AI models.

2.3 Non-Functional Requirements

The application should provide fast response times, high availability, simple navigation, security, maintainability, scalability, and compatibility with modern web browsers. The architecture should support future enhancements without major redesign.

2.4 Hardware Requirements

Processor:	Intel	Core	i3	or	above
RAM:	Minimum	4	GB	(8	GB recommended)
Storage:	1	GB		free	space

Internet connection for AI services.

2.5 Software Requirements

Operating	System:	Windows	10/11
Programming	Language:	Python	3.x
Backend:			FastAPI
Frontend:	HTML,	CSS,	Jinja2
AI Models:	Google	Gemini	API and LaMini-Flan-T5
IDE:	Visual	Studio	Code

Version Control: Git and GitHub.

2.6 Feasibility Study

Technical feasibility is achieved through open-source technologies and cloud AI APIs. Economic feasibility is ensured because the project uses free development tools. Operational feasibility is high due to its simple interface and educational focus. Schedule feasibility is achieved through modular development.

2.7 System Constraints

The application depends on internet connectivity for cloud AI services. API limits, server performance, and browser compatibility may affect performance. Future improvements can reduce these limitations.

2.8 Chapter Summary

This chapter described the functional requirements, non-functional requirements, hardware and software requirements, feasibility study, and system constraints that guided the development of the EduGenie Learning Assistant.